

Can AI detect masks in real time?

A Deep Learning Project



Using Computer Vision to build a **safer & smarter** world.



Image /
Camera Input



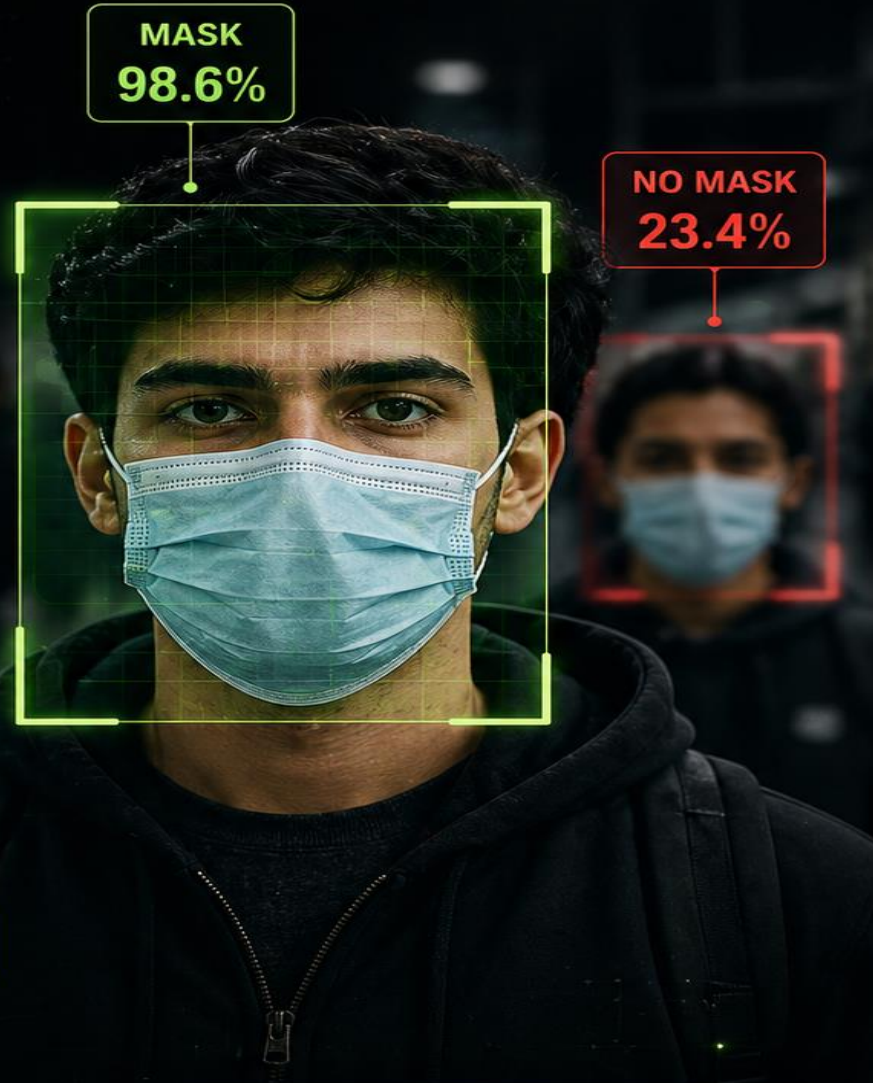
Deep Learning
Model (CNN)



Prediction
+ Confidence



Real-time
Detection



AI today. **Safer tomorrow.**



THE PROBLEM

Monitoring mask usage manually is **difficult**.

In public places, offices, and transport hubs, it's nearly impossible to continuously monitor everyone.

WHY THIS MATTERS?



Protect public health
Reduce risk of virus transmission.



Save time & resources
Automate monitoring with AI.



Ensure compliance
Promote a safer environment.



High footfall,
hard to monitor



Manual checks are
time-consuming



Not scalable for
offices & large spaces



Non-compliance leads
to health risks



THE SOLUTION

Built a **CNN model** that detects:



MASK

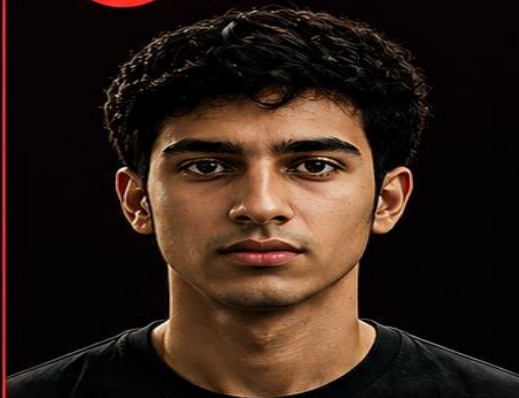


98.6%

Confidence



NO MASK



23.4%

Confidence



CNN MODEL



The model classifies faces into two categories with **high accuracy**.



HOW IT WORKS

A simple pipeline from **image** to **prediction**

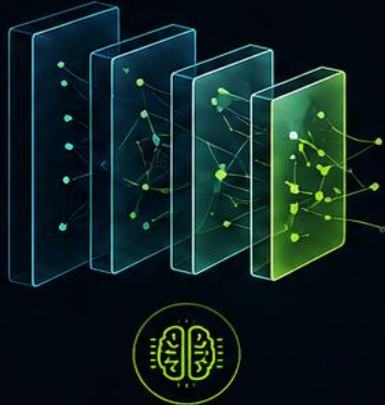
1 IMAGE INPUT



Upload or capture an image using the camera.



2 MODEL PROCESSING



CNN model analyzes the image and extracts important features.



3 PREDICTION



The model predicts whether the person is wearing a mask or not.



4 CONFIDENCE SCORE



Displays the prediction result along with the confidence score.



Powered by **Deep Learning** to deliver accurate and real-time results.



High Accuracy
Trained on a large and diverse dataset.



Real-time
Fast inference for instant results.



Reliable
Built for safety, privacy & trust.



TECH STACK

Built using **powerful tools**
and frameworks.



PYTHON

Core programming
language



TENSORFLOW / KERAS

Build and train the
CNN model



STREAMLIT

Create interactive web app
for real-time predictions



This tech stack enables **fast development**, **accurate predictions**,
and an **interactive user experience**.



REAL-WORLD USE CASES

AI-powered mask detection can make our public spaces **safer**.



AIRPORTS

Enhance safety by ensuring mask compliance.



Automate entry screening and reduce manual monitoring.



OFFICES

Automated entry monitoring for employees & visitors.



Helps maintain a safe and healthy work environment.



MALLS & RETAIL

Ensure customer safety in crowded areas.



Promote responsible behavior and build customer trust.



From public safety to workplaces,
AI makes compliance smarter and faster.



KEY LEARNINGS FROM THIS PROJECT

*Building the model was just one part,
making it usable is what **creates impact**.*



Problem Solving

Identified a real-world problem and built an AI solution.



Deep Learning

Learned and implemented CNN for image classification.



End-to-End Development

From data preprocessing to model training and real-time inference.



UI/UX Matters

Built an interactive interface using Streamlit for real-world usability.



This project strengthened my skills in **Deep Learning**, **Computer Vision** and building real-world AI applications.



AI MODEL IS POWERFUL,
BUT UI MAKES IT USABLE.





App Interface

Navigation

Go to

- Home
- How It Works
- Concepts Used
- Prediction
- Examples & Use Cases
- FAQ / Common Issues



Face Mask Detection App

Deploy



Project Overview

This Deep Learning application detects whether a person is **wearing a face mask or not** using CNN.

Technologies Used

- Python
- TensorFlow / Keras
- CNN
- Streamlit



Uploaded Image

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Input Image

Predict

✓ Prediction: With Mask

Confidence: 95.01%



With Mask (95%)



Supports both
uploaded
images &
real-time
camera input



Camera Capture

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Input Image

Predict

✗ Prediction: Without Mask

Confidence: 99.51%



Without Mask (99%)



Real-time Prediction

Instant results with high speed



High Accuracy

Trained CNN model for reliable
mask detection



User-Friendly UI

Built with Streamlit for a smooth
and interactive experience



github.com/ganesh838/

Ganesh

| AI Projects

